BUILDING A RETAIL SALES DATA AGENT ON DATABRICKS SUBMISSION

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Introduction & Objectives

We have been instructed to build a data agent using Databricks. We were given a data set CSV file containing data that was uploaded on Databricks to create the table retail sales data containing 1000 rows or transactions.

The purpose of the agent is to allow staff members to ask questions using straight language about sales performance and receive accurate answers.

The objective was to build an end-to-end data agent: upload the data, prepare the table, write original agent instructions, test the agent with real business questions, validate its answers, and document the entire process.

Dataset Descriptions

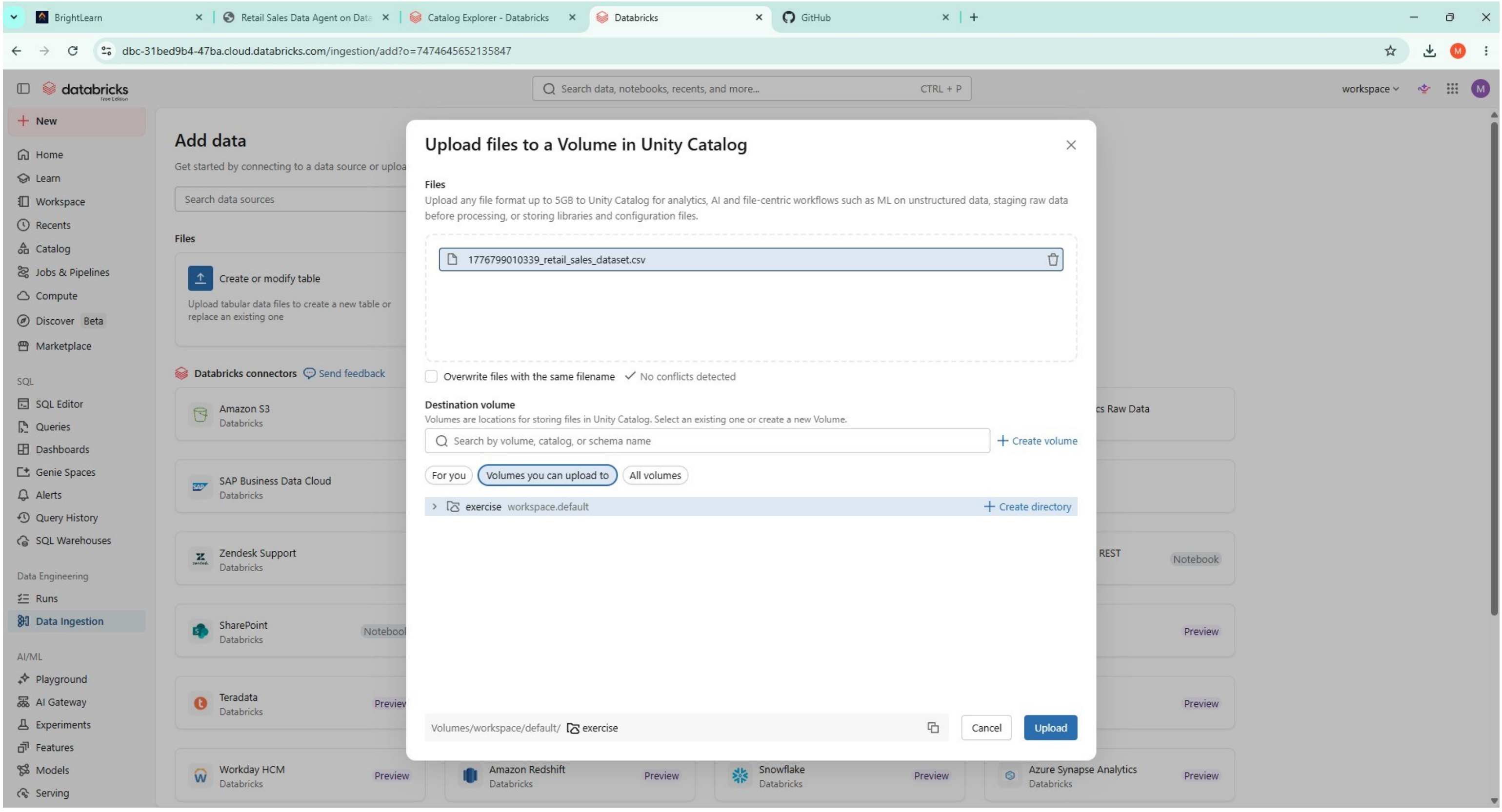
The dataset covers three product categories: Electronics, Clothing, and Beauty. It includes customer information such as gender and age, making it possible to analyse both product performance and customer behaviour patterns.

The dataset used in this project is the retail_sales_data table stored in Databricks (workspace.default.retail_sales_data). It contains 1,000 rows of retail sales transactions from January 2023 to January 2024.

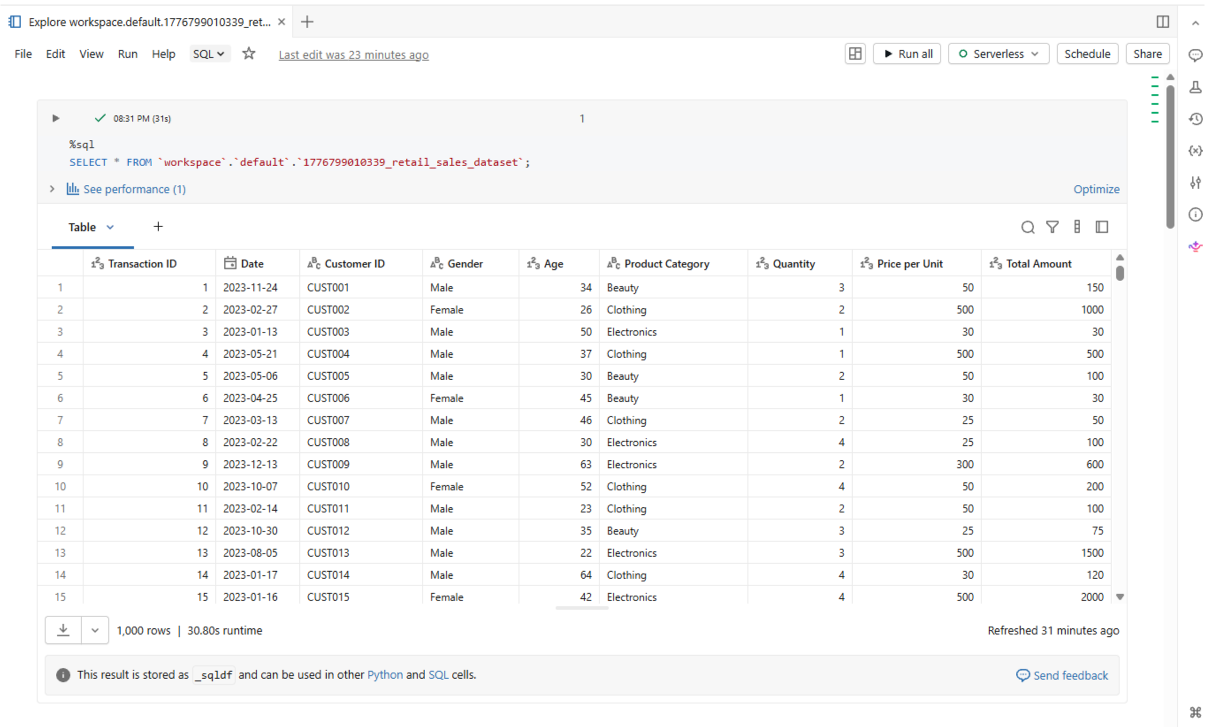
Column Name	Data Type	Description
transaction_id	Integer	ID for each sale, numbered 1 to 1000
date	Date	Date the sale occurred in YYYY-MM-DD format
customer_id	String	Customer identifier
gender	String	Gender of the customer: Male or Female only
age	Integer	Age of the customer in years, between 18 and 64
product_category	String	Type of product: Electronics, Clothing or Beauty
quantity	Integer	Number of units purchased in the transaction
price_per_unit	Integer	Price of a single unit in dollars
total_amount	Integer	Total value of the transaction

1. **Upload:**

   Upload 1 Failed – only uploaded 50 rows. Opted for option 2 for upload which uploaded all rows. See screenshot.



2. **Review of the Dataset**



	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
7	7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
8	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
9	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
10	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
11	11	2023-02-14	CUST011	Male	23	Clothing	2	50	100
12	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
13	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
14	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
15	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000

Workspace

Tree view: ON

.../ mongaumatso...

Type to search

Drafts

Explore workspace.default.1776799010...

Explore workspace.default.1776799010339_ret...

File Edit View Run Help SQL Last edit was 24 minutes ago Run all Serverless

08:38 PM (4s) 2

```sql
SELECT
  COUNT(*) AS total_rows,
  SUM(`Total Amount`) AS total_revenue,
  MIN(Date) AS start_date,
  MAX(Date) AS end_date
FROM workspace.default.`1776799010339_retail_sales_dataset`;
```

See performance (1)

Table

	total_rows	total_revenue	start_date	end_date
1	1000	456000	2023-01-01	2024-01-01

1 row | 3.51s runtime Refresh

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

08:39 PM (2s) 3

```sql
DESCRIBE workspace.default.`1776799010339_retail_sales_dataset`;
```

See performance (1) Optimize

Table

	col_name	data_type	comment
1	Transaction ID	bigint	null
2	Date	date	null
3	Customer ID	string	null
4	Gender	string	null
5	Age	bigint	null
6	Product Category	string	null
7	Quantity	bigint	null
8	Price per Unit	bigint	null
9	Total Amount	bigint	null

9 rows | 2.36s runtime Refreshed 26 minutes ago

08:39 PM (3s) 4

```sql
SELECT
  `Product Category`,
  COUNT(*) AS transactions,
  SUM(`Total Amount`) AS total_revenue
FROM workspace.default.`1776799010339_retail_sales_dataset`
GROUP BY `Product Category`
ORDER BY total_revenue DESC;
```

See performance (1) Optimize

Table

	Product Category	transactions	total_revenue
1	Electronics	342	156905
2	Clothing	351	155580
3	Beauty	307	143515

3 rows | 3.18s runtime Refreshed 26 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

08:40 PM (3s) 6

```sql
SELECT
  MIN(Age) AS youngest,
  MAX(Age) AS oldest
FROM workspace.default.`1776799010339_retail_sales_dataset`;
```

See performance (1) Optimize

Table +

	youngest	oldest
1	18	64

1 row | 3.43s runtime Refreshed 27 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

08:39 PM (3s) 5

```sql
SELECT
  Gender,
  COUNT(*) AS customers,
  SUM(`Total Amount`) AS total_revenue
FROM workspace.default.`1776799010339_retail_sales_dataset`
GROUP BY Gender;
```

See performance (1) Optimize

Table +

	Gender	customers	total_revenue
1	Male	490	223160
2	Female	510	232840

2 rows | 2.89s runtime Refreshed 26 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

3. Prepare the table

2 minutes ago (18s) 7

```sql
CREATE OR REPLACE TABLE workspace.default.retail_sales_data AS
SELECT
  `Transaction ID` AS transaction_id,
  Date AS date,
  `Customer ID` AS customer_id,
  Gender AS gender,
  Age AS age,
  `Product Category` AS product_category,
  Quantity AS quantity,
  `Price per Unit` AS price_per_unit,
  `Total Amount` AS total_amount
FROM workspace.default.`1776799010339_retail_sales_dataset`;
```

See performance (1) Optimize

Table +

num_affected_rows	num_inserted_rows

No rows returned

0 rows | 18.06s runtime Refreshed 2 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

09:20 PM (6s) 8

```sql
SELECT COUNT(*) AS total_rows
FROM workspace.default.retail_sales_data;
```

See performance (1) Optimize

Table +

	total_rows
1	1000

1 row | 6.14s runtime Refreshed 5 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

09:21 PM (3s) 9 SQL

```sql
DESCRIBE TABLE EXTENDED workspace.default.retail_sales_data;
```

See performance (1) Optimize

Table +

	col_name	data_type	comment
1	transaction_id	bigint	null
2	date	date	null
3	customer_id	string	null
4	gender	string	null
5	age	bigint	null
6	product_category	string	null
7	quantity	bigint	null
8	price_per_unit	bigint	null
9	total_amount	bigint	null
10			
11	# Delta Statistics Columns		
12	Column Names	quantity, customer_id, price_per_unit, age, transaction_id, date, product_category, total_amount, gender	
13	Column Selection Method	first-32	
14			
15	# Detailed Table Information		

29 rows | 2.79s runtime Refreshed 5 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

Search data, notebooks, recents, and more... CTRL + P workspace

Catalog Focus: ON

Serverless Starter Warehouse Serverless 2XS

Catalog

Type to search

- My organization
  - workspace
    - default
      - Tables (2)
        - 1776799010339_retail_sales_dataset
        - retail_sales_data
      - Volumes (1)
        - exercise
    - information_schema
  - system
- Delta Shares received
  - samples

Catalog Explorer › workspace › default ›

retail_sales_data

Open in a dashboard Share Create

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Description

AI generate Add

Filter columns... AI generate

Column	Type	Comment	Tags	Column masking
transaction_id	bigint			
date	date			
customer_id	string			
gender	string			
age	bigint			
product_category	string			
quantity	bigint			
price_per_unit	bigint			
total_amount	bigint			

About this table

Owner mongaumatsobane@gmail.com

Type Managed

Data source Delta

Popularity

Quality

Not enabled

Data Quality Monitoring is not enabled for this table

Enable

Tags

Add tags

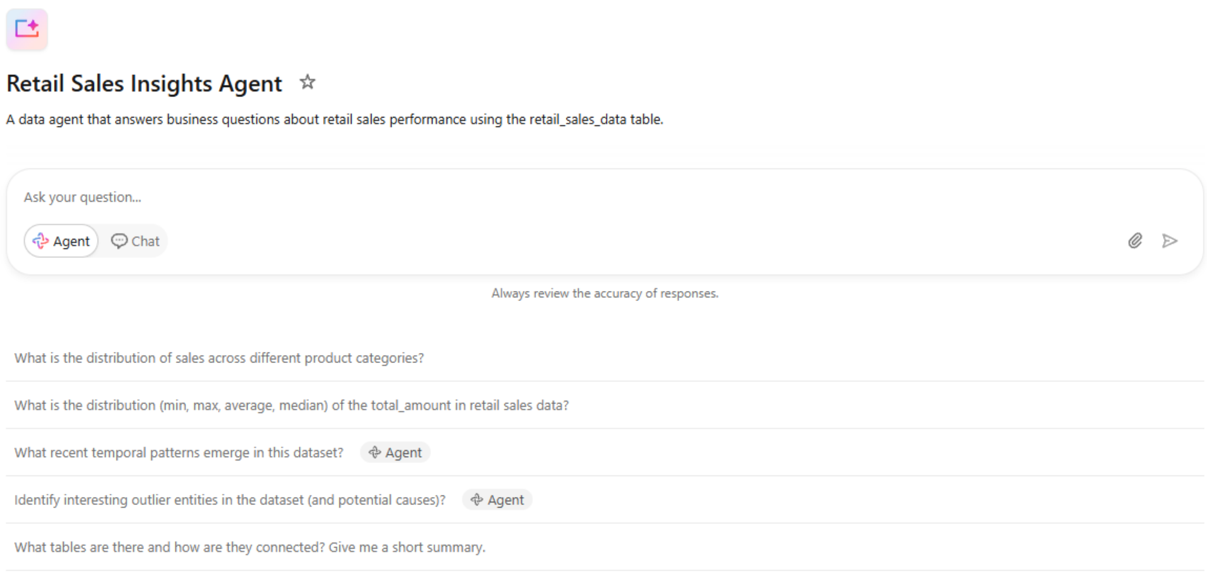
Row filter

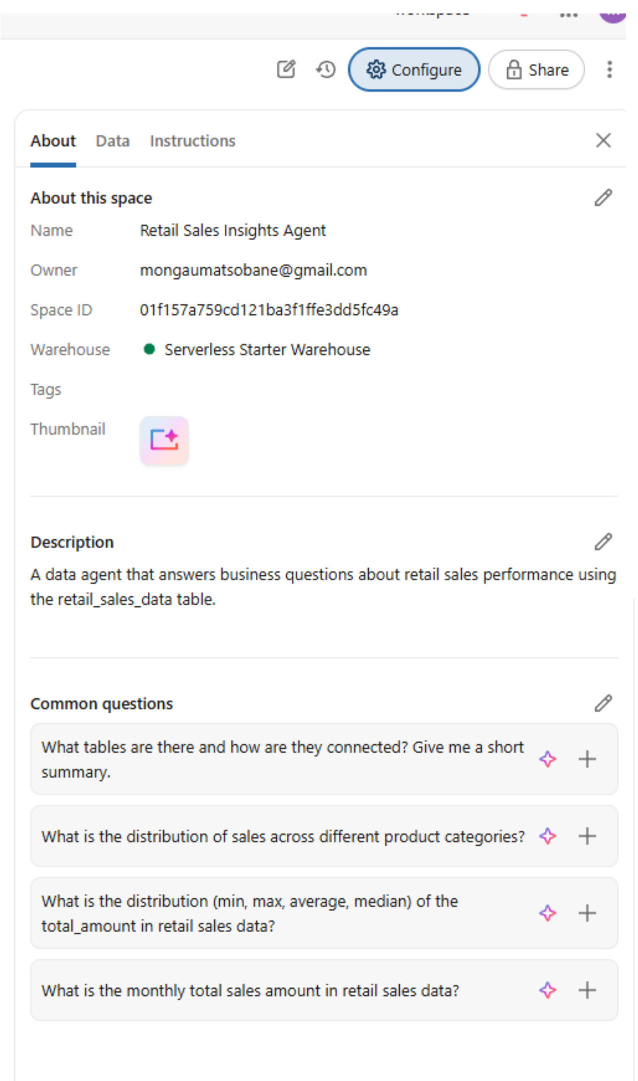
Add filter

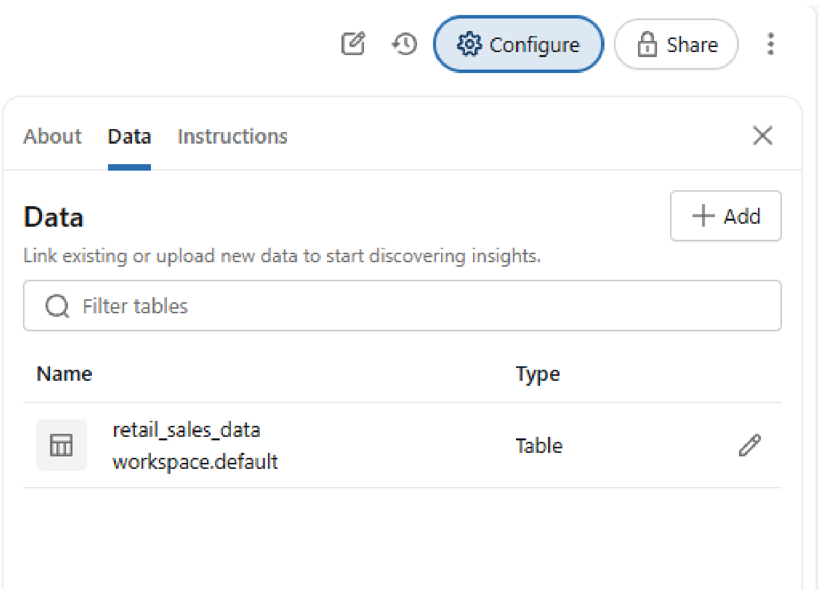
Policies

New policy

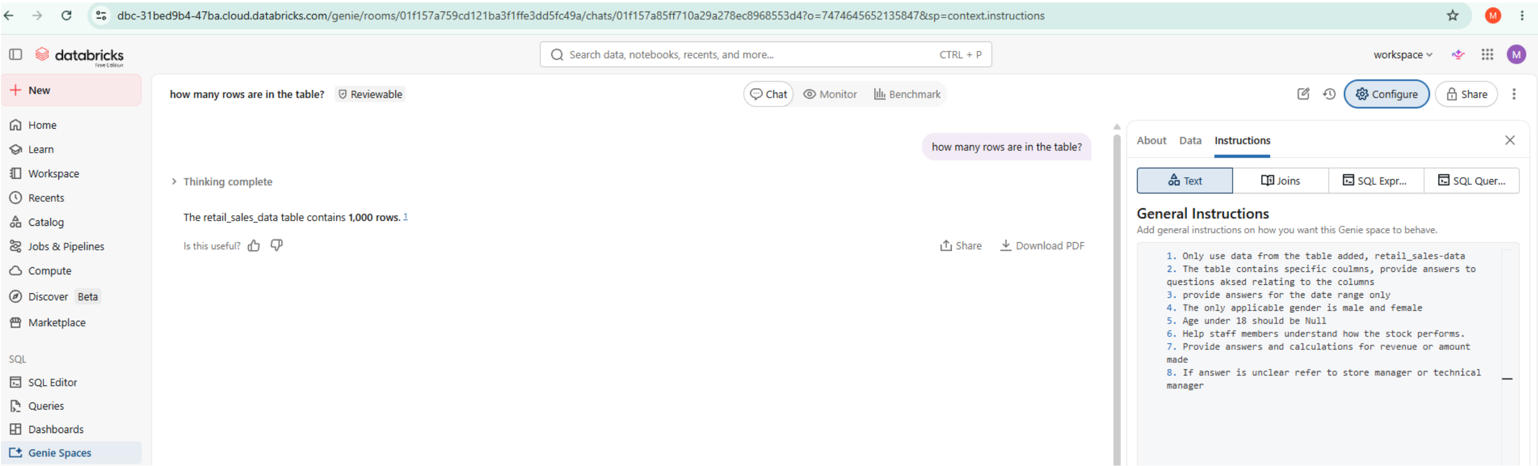
4. Create the Data Agent

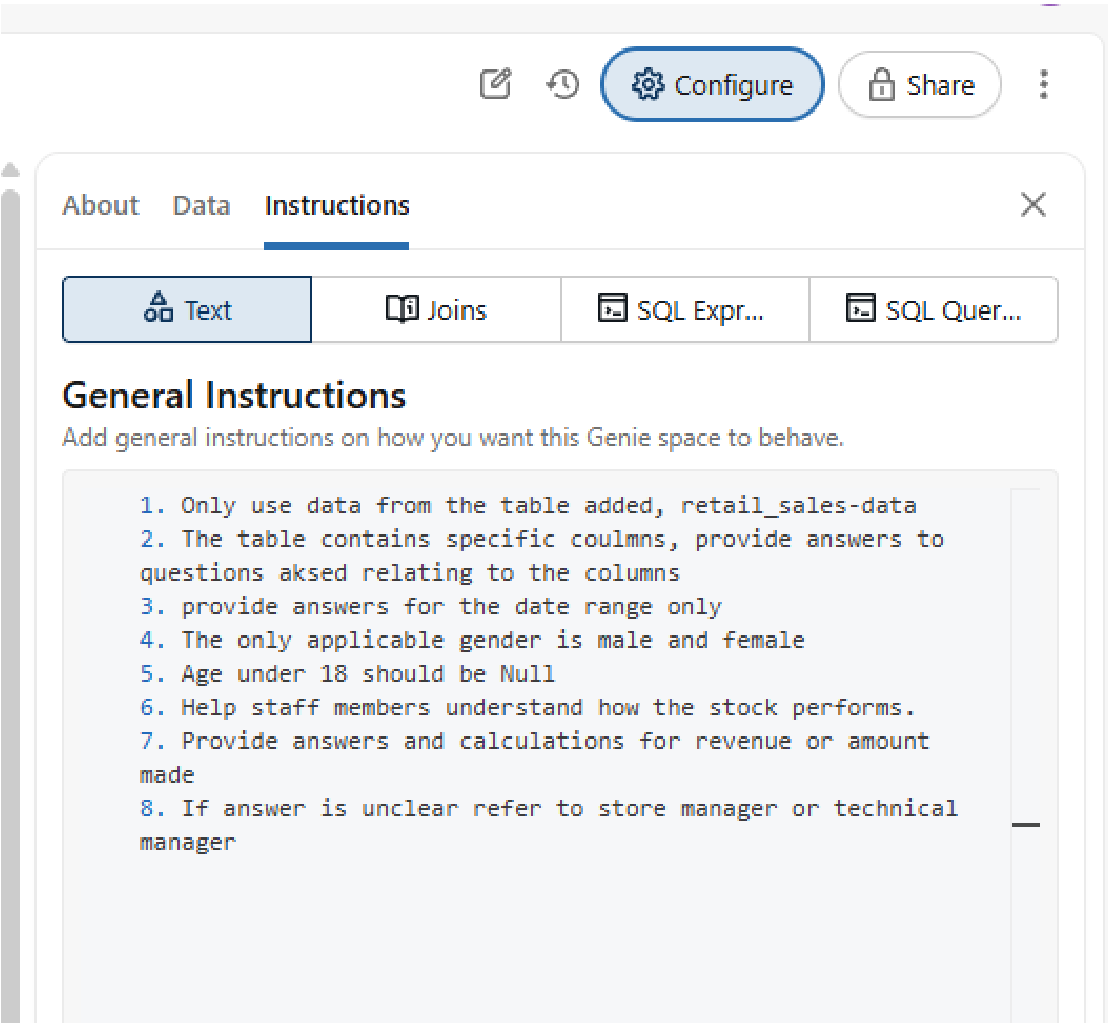






5. Own Instructions





6. Test with 10 Questions

what is the total amount made in 2023 between January and March?

Thinking complete

The total amount made between January and March 2023 was **$108,500.** [2]

Is this useful?

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Which product was bought the most?

Thinking complete

Clothing was the most purchased product category, with **894 total units sold** across 351 transactions. [3]

Electronics came in second with 849 units sold, followed by Beauty with 771 units.

Is this useful?

Share Download PDF

How much did the most bought product make?

Thinking complete

Clothing, the most purchased product category, generated **$155,580** in total revenue. [4]

Is this useful?

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How much did beauty products make

Thinking complete

Beauty products generated **$143,515** in total revenue. [5]

Is this useful?

Share Download PDF

name all beauty products

Thinking complete

The table does not contain individual product names. It only tracks transactions by **product category** (Beauty, Clothing, or Electronics), not specific product names. [6]

The available information for Beauty products includes:

- Transaction details (date, customer, quantity)
- Price per unit
- Total amount per transaction

If you need information about specific beauty product names or SKUs, please refer to the store manager or technical manager, as this data is not available in the current sales table.

Is this useful?

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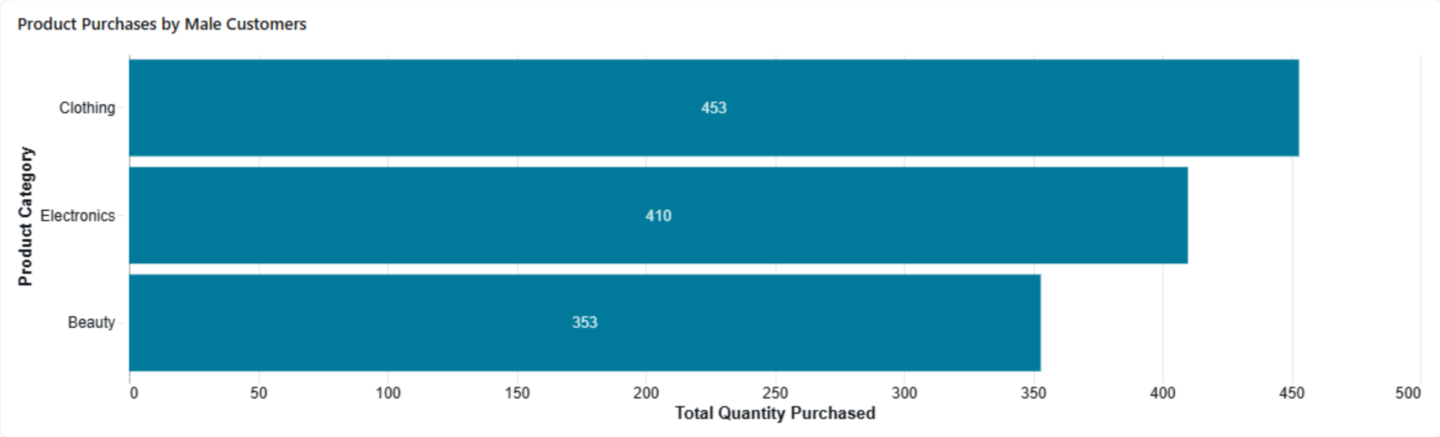
which products do male customers buy the most?

Thinking complete

Male customers buy **Clothing** the most, with **453 total units purchased** across 177 transactions. 7



Source: Product purchases by male customers

Breakdown by Product Category

Male customers' purchasing patterns show:

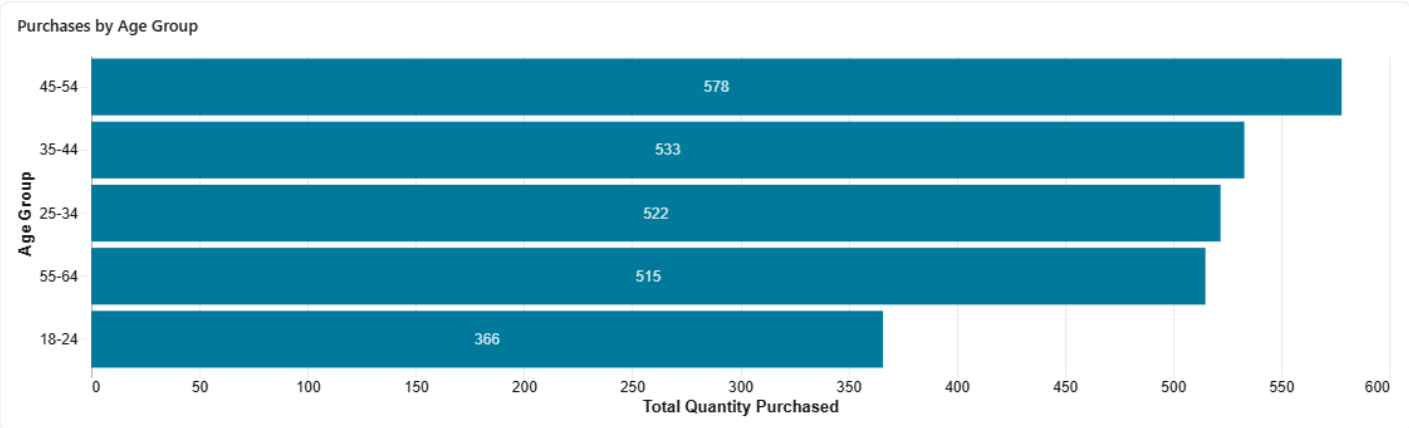
- **Clothing:** 453 units purchased, generating $74,305 in revenue
- **Electronics:** 410 units purchased, generating $80,170 in revenue
- **Beauty:** 353 units purchased, generating $68,685 in revenue

While Clothing leads in quantity purchased, it's worth noting that Electronics generated the highest revenue among male customers despite having fewer units sold, suggesting higher average prices for electronics products.

Which age group buys the most?

Thinking complete

The **45-54 age group** buys the most, with **578 total units purchased** across 225 transactions. 9



Source: Purchases by age group

Age Group Breakdown

The purchasing patterns across age groups show:

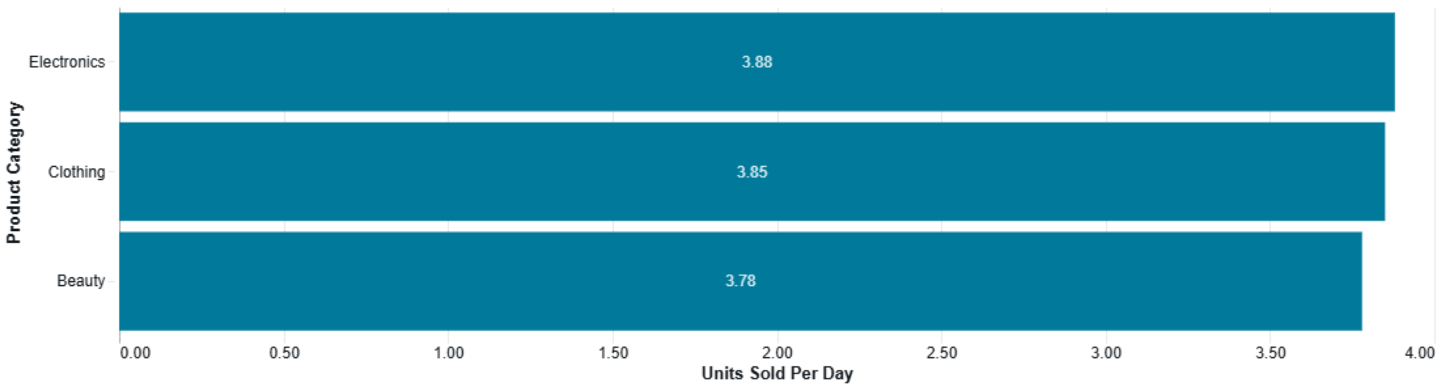
1. **45-54 years:** 578 units purchased, $97,235 in revenue (highest quantity)
2. **35-44 years:** 533 units purchased, $96,835 in revenue
3. **25-34 years:** 522 units purchased, $97,090 in revenue
4. **55-64 years:** 515 units purchased, $90,190 in revenue
5. **18-24 years:** 366 units purchased, $74,650 in revenue (lowest quantity)

The middle-aged customers (35-54 years) represent the strongest purchasing segment, accounting for over 1,100 units combined. The youngest age group (18-24) shows notably lower purchasing which product should be restocked often

› Thinking complete

Electronics and Clothing should be restocked most often, as they have the highest sales velocity at **3.88 and 3.85 units per day** respectively. [12,14]

Daily Sales Velocity by Product Category



Source: Daily sales velocity for visualization

Restocking Priority Analysis

Based on sales velocity and turnover rates, here's the restocking priority:

Restocking Priority Summary

Product Category	Total Units Sold	Units Per Day	Transactions Per Day	Restocking Priority
Electronics	849	3.88	1.56	High Priority
Clothing	894	3.85	1.51	High Priority
Beauty	771	3.78	1.50	Medium Priority

Source: Restocking priority by turnover rate

Key Findings

High Priority Products:

- **Electronics:** 3.88 units/day, 1.56 transactions/day - Highest daily velocity despite having 219 active sales days
- **Clothing:** 3.85 units/day, 1.51 transactions/day - Most total units sold (894) with consistent demand across 232 days

Medium Priority:

- **Beauty:** 3.78 units/day, 1.50 transactions/day - Slightly lower velocity but still steady demand

Seasonal Demand Patterns

The data reveals important seasonal trends that should inform restocking schedules: [13]

- **Clothing:** Peak demand in March (111 units) and May (97 units), lowest in July (45 units) - consider increasing stock before spring season
- **Electronics:** Strong demand in May (97 units), August (87 units), October (95 units), and December (92 units) - likely driven by holiday shopping
- **Beauty:** Most consistent demand throughout the year, with a peak in October (83 units)

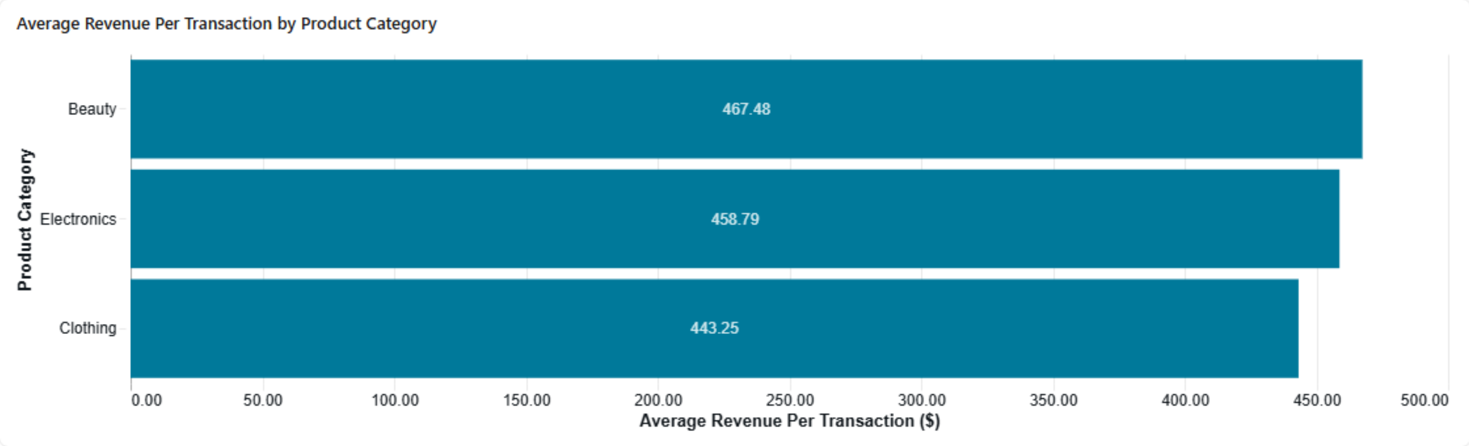
Recommendation

Focus restocking efforts on **Electronics and Clothing** with more frequent inventory checks. Electronics should be prioritized during May, August, October, and December, while Clothing requires extra attention during spring months (March-May).

Is this useful?   Share   Download PDF what is the average made per each product category?

Thinking complete

Beauty products have the highest average revenue per transaction at **$467.48**, followed by Electronics at $458.79 and Clothing at $443.25. [16]



Source: Average revenue per transaction by product category

Average Revenue Breakdown

Average Revenue Summary by Product Category

Product Category	Total Revenue	Transactions	Avg Per Transaction	Avg Price Per Unit	Avg Quantity
Beauty	$143,515	307	$467.48	$184.06	2.51
Electronics	$156,905	342	$458.79	$181.90	2.48
Clothing	$155,580	351	$443.25	$174.29	2.55

Source: Average revenue per transaction by product category

Key Insights

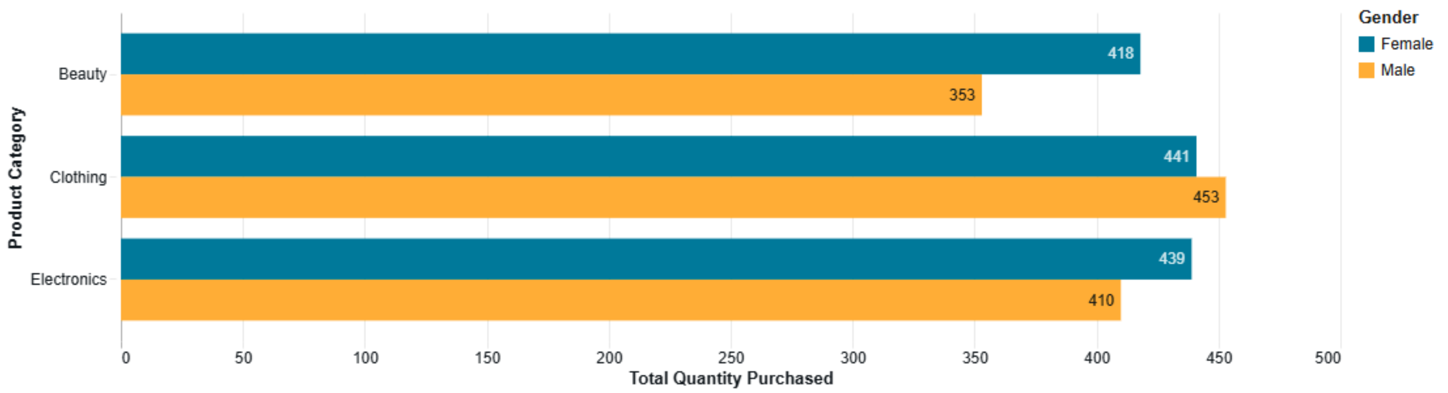
- **Beauty:** $467.48 average per transaction, with an average price of $184.06 per unit and 2.51 units per transaction
- **Electronics:** $458.79 average per transaction, with an average price of $181.90 per unit and 2.48 units per transaction
- **Clothing:** $443.25 average per transaction, with an average price of $174.29 per unit and 2.55 units per transaction

While Clothing has the highest sales volume (894 units total), Beauty products generate the highest average transaction value, making them the most valuable per sale. Electronics also command strong per-transaction revenue despite having the lowest average quantity per purchase.

Which products are bought more by females?

› Thinking complete

Beauty products are bought more by females, representing **54.22%** of all Beauty purchases, followed by Electronics at 51.71%. [18]



Source: Product quantity by category and gender for visualization

Gender Preference Analysis

Female Purchase Preference by Product Category

Product Category	Female Purchases	Male Purchases	Female %
Beauty	418	353	54.22
Electronics	439	410	51.71
Clothing	441	453	49.33

Source: Gender breakdown by product category

Key Findings

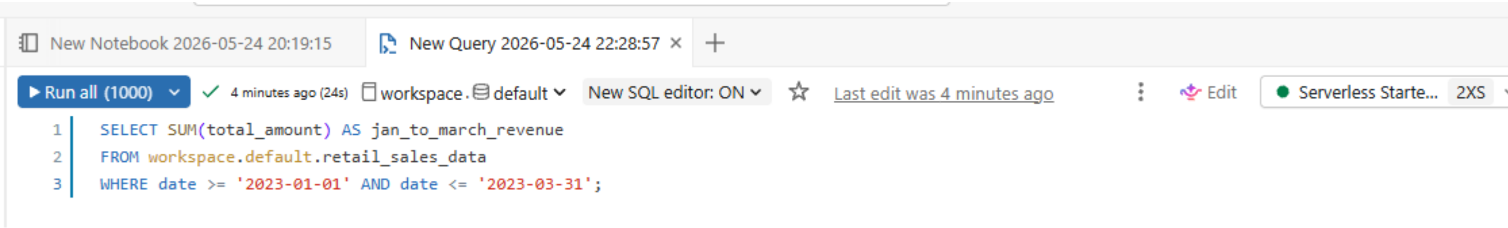
Female customers show a clear preference for:

1. **Beauty:** 418 units purchased by females vs. 353 by males (54.22% female) - the strongest female preference
2. **Electronics:** 439 units purchased by females vs. 410 by males (51.71% female) - slight female preference
3. **Clothing:** 441 units purchased by females vs. 453 by males (49.33% female) - actually slightly more popular with males

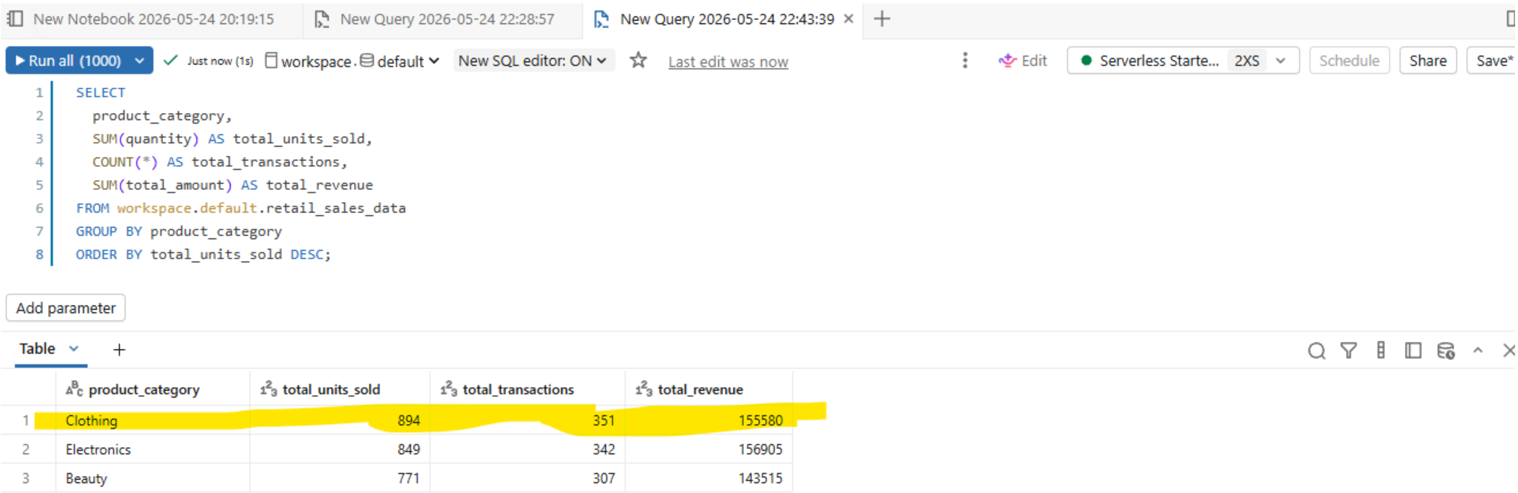
While female customers purchase **Clothing** in the highest absolute quantity (441 units), Beauty products show the strongest female preference when comparing the gender split. Interestingly, Clothing is the only category where male customers actually purchase slightly more than females.

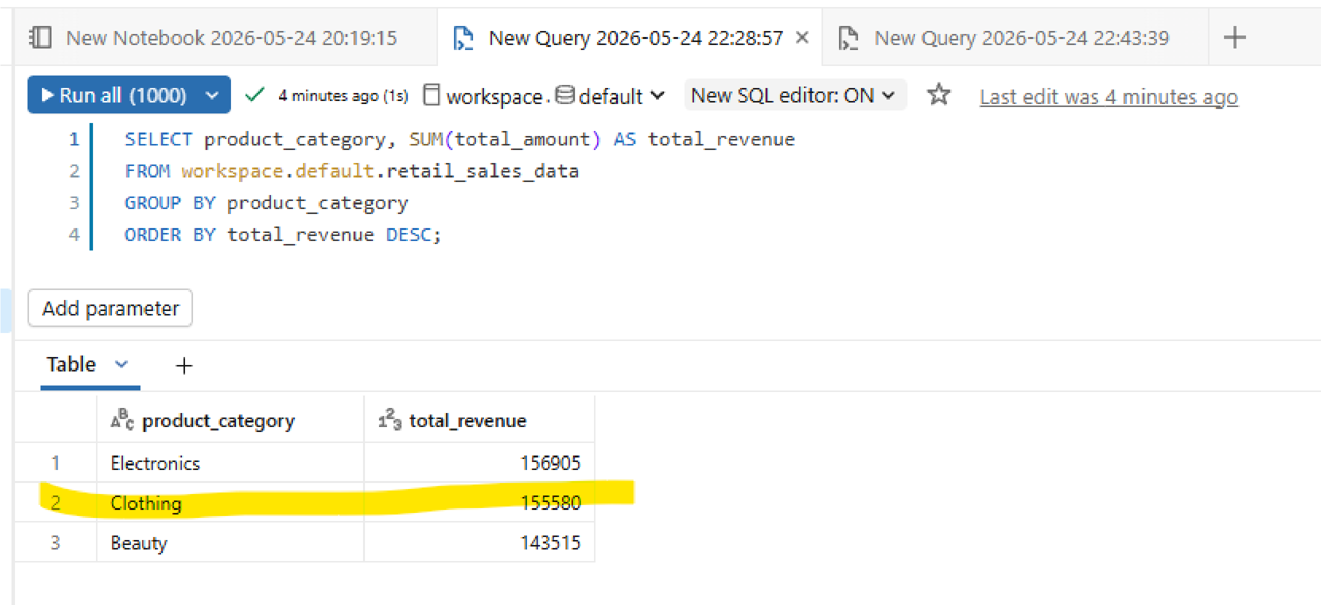
7. **Validate the Answers**

- Validation Question: What is the total amount made between January and March 2023?
- Agent Answer: $108,500
- SQL Query Answer: $108,500
- Answer is correct.



- Validation Question: How much did the most bought product make? (Clothing)
- Agent Answer: $155,580
- SQL Query Answer: 155580
- Answer is correct.





- Validation Question: Which age group buys the most?
- Agent Answer: 45-54
- SQL Query Answer: 45-54
- Answer is correct.

New Notebook 2026-05-24 20:19:15 | New Query 2026-05-24 22:28:57 | New Que

Run all (1000) Just now (1s) workspace . default New SQL editor: ON

```sql
SELECT
  CASE
    WHEN age BETWEEN 18 AND 24 THEN '18-24'
    WHEN age BETWEEN 25 AND 34 THEN '25-34'
    WHEN age BETWEEN 35 AND 44 THEN '35-44'
    WHEN age BETWEEN 45 AND 54 THEN '45-54'
    WHEN age BETWEEN 55 AND 64 THEN '55-64'
  END AS age_group,
  SUM(quantity) AS total_units,
  SUM(total_amount) AS total_revenue
FROM workspace.default.retail_sales_data
GROUP BY age_group
ORDER BY total_units DESC;
```

Add parameter

Table

	age_group	total_units	total_revenue
1	45-54	578	97235
2	35-44	533	96835
3	25-34	522	97090
4	55-64	515	90190
5	18-24	366	74650